

$$1. A = \{\pi, e, 0\}, B = \{0, 1\}$$

$$(a) A \times B = \{0, \pi, 0, e, 0, 0\} \triangle (b) B \times A = \{0, 0, 0, \pi, e, 0\} \triangle (c) A \times A = \{\pi^2, \pi e, 0, e\pi, e^2, 0, 0, 0, 0\}$$

$$(d) B \times B = \{0, 0, 0, 1\} \triangle (e) \{\}$$